Lec5

Matrix Decomposition

CS 556

Erisa Terolli

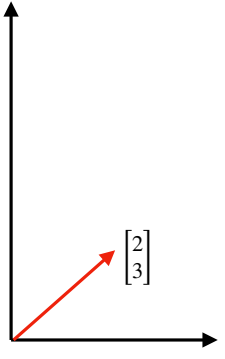
Outline

1. Eigen Decomposition
2. Singular Value Decomposition (SVD)
3. Dimensionality Reduction with SVD
4. Movie Recommender System with SVD

2D Transformation Matrix

What happens when we multiply a 2D matrix with a vector?

$$\begin{bmatrix} 2 & -1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} 2 \\ 3 \end{bmatrix} = ?$$

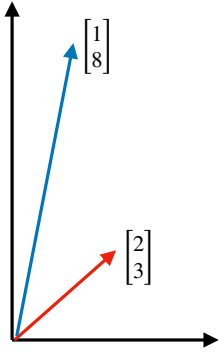




2D Transformation Matrix

What happens when we multiply a 2D matrix with a vector?

$$\begin{bmatrix} 2 & -1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} 2 \\ 3 \end{bmatrix} = \begin{bmatrix} 1 \\ 8 \end{bmatrix}$$

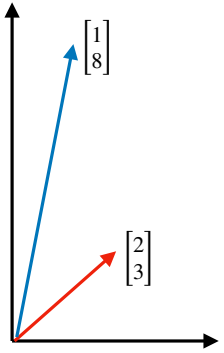


1.3

2D Transformation Matrix

What happens when we multiply a 2D matrix with a vector?

$$\begin{bmatrix} 2 & -1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} 2 \\ 3 \end{bmatrix} = \begin{bmatrix} 1 \\ 8 \end{bmatrix}$$



The matrix will transform the vector by rotating and stretching/shortening it

$$R = \begin{bmatrix} cos\theta & -sin\theta \\ sin\theta & cos\theta \end{bmatrix}$$

Rotation by θ

$$S = \begin{bmatrix} \alpha & 0 \\ 0 & \alpha \end{bmatrix}$$

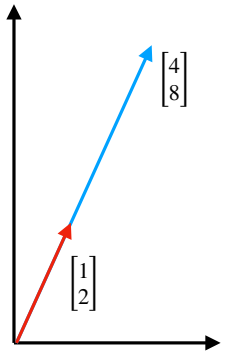
Stretching by α



2D Transformation Matrix

What happens when we multiply a 2D matrix with a vector?

$$\begin{bmatrix} 2 & 1 \\ 2 & 3 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 4 \\ 8 \end{bmatrix}$$



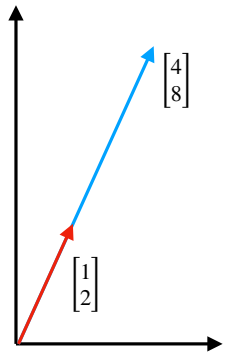
1.5

2D Transformation Matrix

What happens when we multiply a 2D matrix with a vector?

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$$4 \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 4 \\ 8 \end{bmatrix}$$



1.6

2D Transformation Matrix

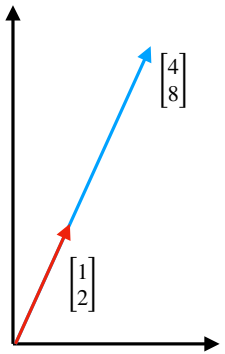
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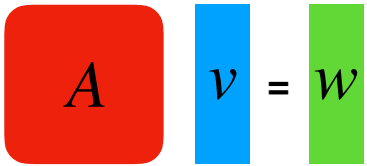
Eigen value

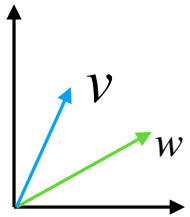
Eigen vector



2.1

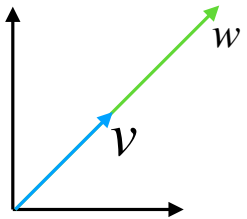
Eigenvectors & Eigenvalues



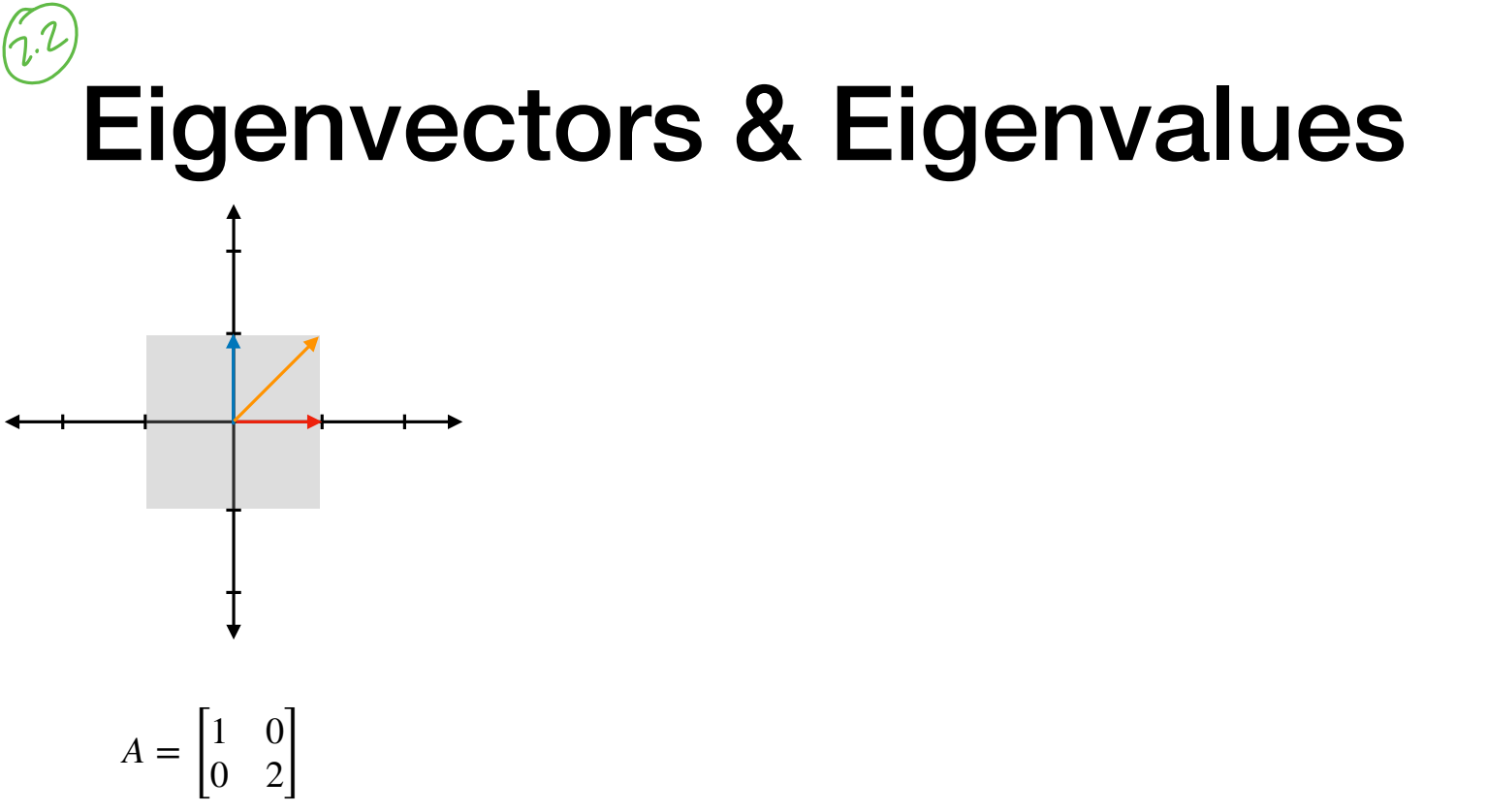


Transformation matrix A is applied to a vector v and outputs a vector w. If w points in the same direction as v (a.k.a. lies on the same 1 dimensional subspace), then v is an eigenvector of matrix A.

$$Av = w$$
$$\lambda v = w$$
$$Av = \lambda v$$



λ is an eigenvalue associated with eigenvector v of A

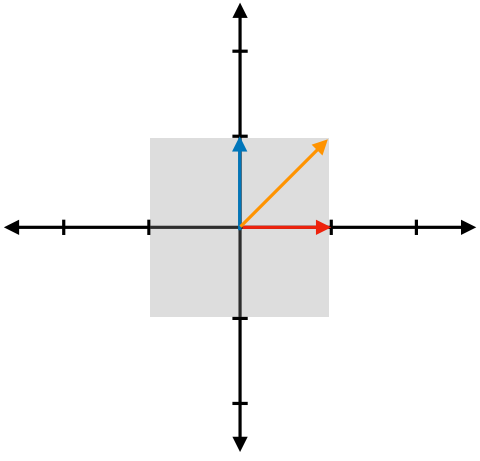


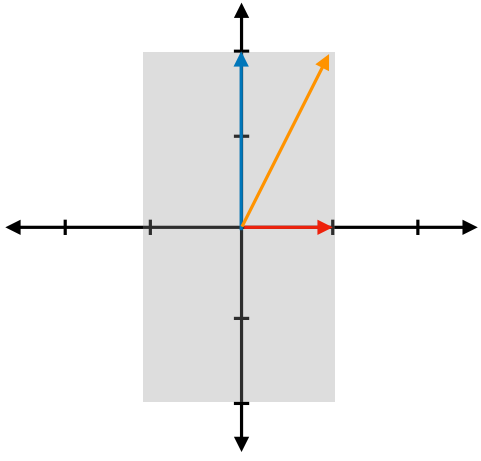
Eigenvectors & Eigenvalues

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}$$

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Eigenvectors & Eigenvalues

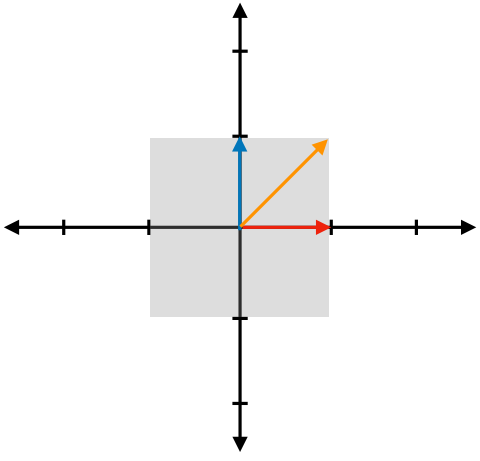


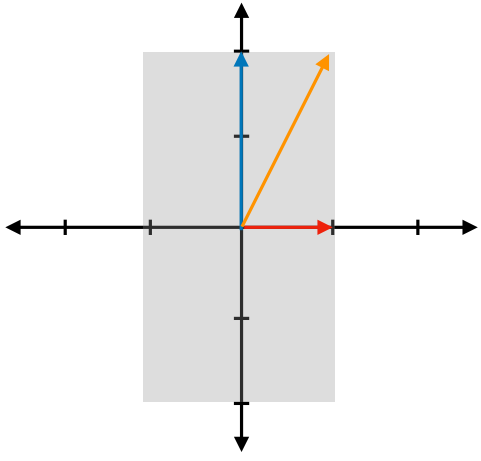


$$A = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}$$

2.4

Eigenvectors & Eigenvalues



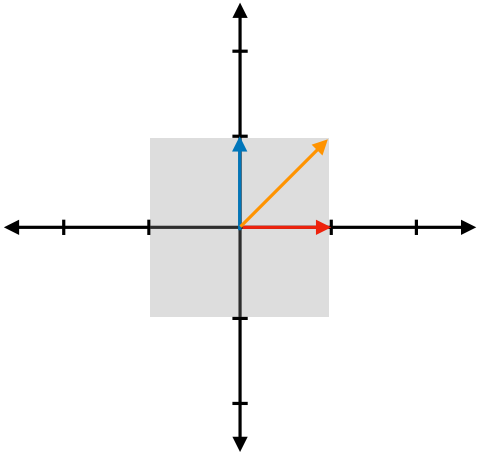


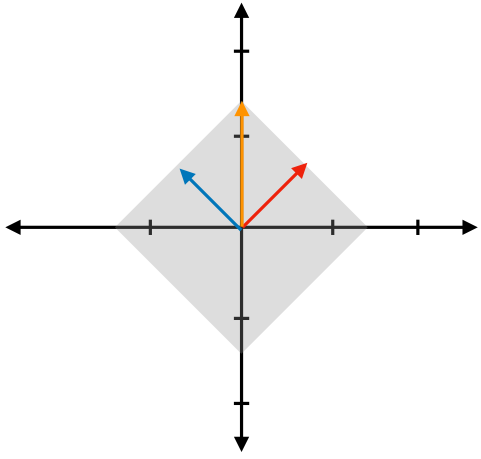
$$A = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}$$

2 eigenvectors with values 1 and 2

2.5

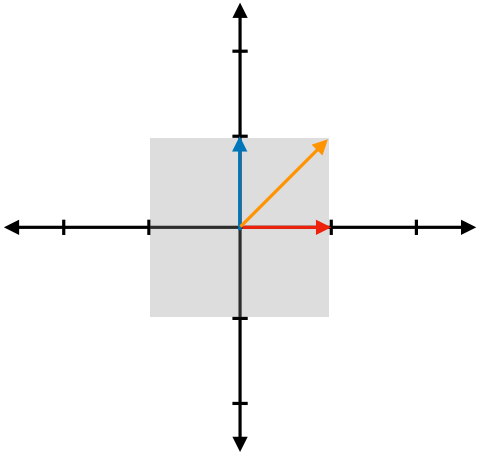
Eigenvectors & Eigenvalues

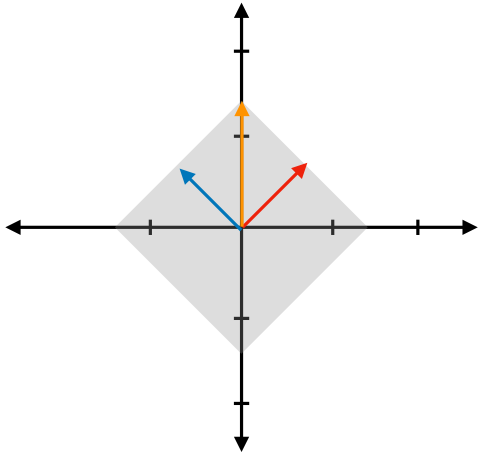




2.6

Eigenvectors & Eigenvalues

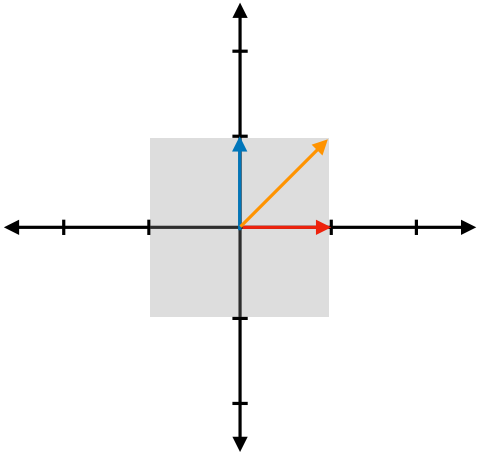


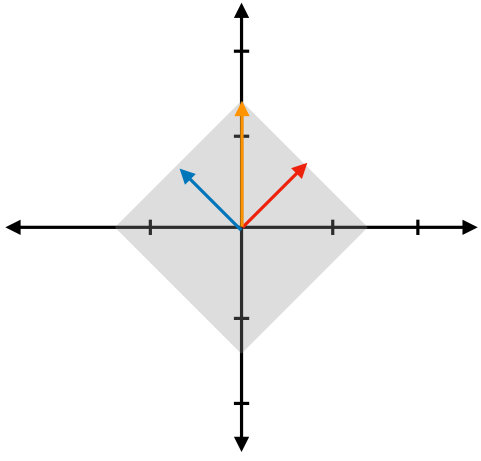


0 Eigenvectors

2.7

Eigenvectors & Eigenvalues





0 Eigenvectors

Check the vectors that lie on the same span after transformation and measure how much they magnitude change

Finding Eigenvalues



2.8

$$Av = \lambda v$$

$$Av - \lambda v = 0$$

$$(A - \lambda)v = 0$$

Shift the matrix A by λ ⇢ $(A - \lambda I)v = 0$

$Ax = 0$
$x \in N(A)$
$v \in N(A - \lambda I)$

It has nontrivial null space. It must be singular. ⇢ $A - \lambda I$

$$det(A - \lambda I) = 0$$ ⇠ The Determinant of singular matrices is 0.

Invertible matrix: how many vectors in the null space.

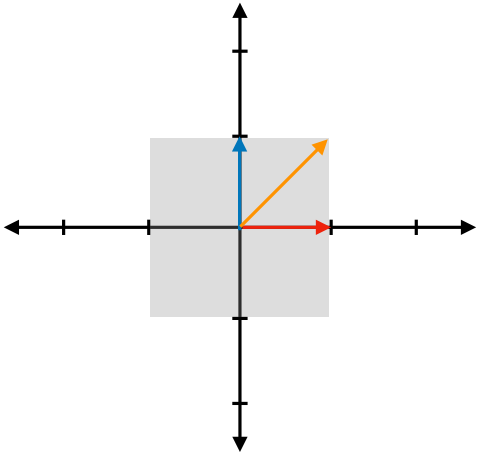
2.9

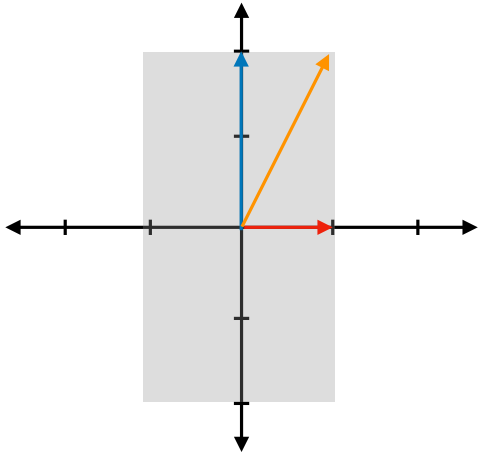
Finding Eigenvectors

1. Find all eigenvalues λ
2. For each λ , find $v \in N(A - \lambda I)$
   - Find a vector v that is in the null space of the matrix $(A - \lambda I)$

2.10

Example 1





$$A = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}$$

1. $$det\left(\begin{bmatrix} 1-\lambda & 0 \\ 0 & 2-\lambda \end{bmatrix}\right) = 0$$

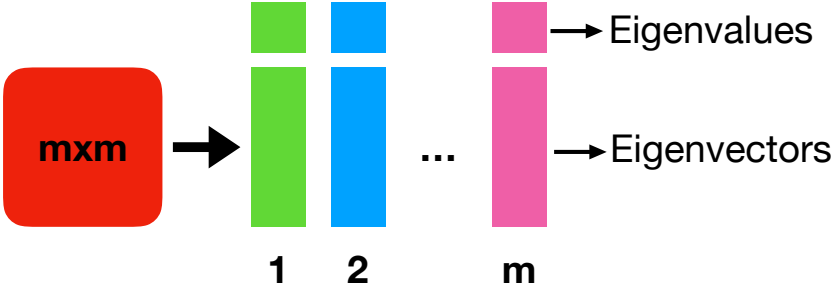
2. $$(1-\lambda)(2-\lambda) = 0$$

3. $$\lambda = 1, \quad \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0 \\ v_2 \end{bmatrix} = 0, \quad v = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

4. $$\lambda = 2, \quad \begin{bmatrix} -1 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} -v_1 \\ 0 \end{bmatrix} = 0, \quad v = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

2.11

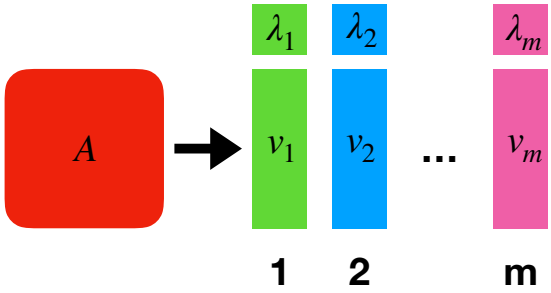
Eigen Decomposition



Eigendecomposition is the factorization of a matrix into a canonical form, whereby the matrix is represented in terms of its eigenvalues and eigenvectors.

Diagonalization

rotating is easier than stretch & rotate.



$$Av_1 = \lambda_1 v_1$$
$$Av_2 = \lambda_2 v_2$$
$$\vdots$$
$$Av_m = \lambda_m v_m$$

must have linearly independent eigenvectors.

(1)

$$\begin{aligned} Av_1 &= \lambda_1 v_1 \\ Av_2 &= \lambda_2 v_2 \\ Av_3 &= \lambda_3 v_3 \end{aligned} \longrightarrow \begin{bmatrix} v_{11} & v_{12} & v_{13} \\ v_{21} & v_{22} & v_{23} \\ v_{31} & v_{32} & v_{33} \end{bmatrix} \begin{bmatrix} \lambda_1 & 0 & 0 \\ 0 & \lambda_2 & 0 \\ 0 & 0 & \lambda_3 \end{bmatrix} = \begin{bmatrix} v_{11}\lambda_1 & v_{12}\lambda_2 & v_{13}\lambda_3 \\ v_{21}\lambda_1 & v_{22}\lambda_2 & v_{23}\lambda_3 \\ v_{31}\lambda_1 & v_{32}\lambda_2 & v_{33}\lambda_3 \end{bmatrix} \longrightarrow AV = V\Lambda$$

(2)

$$AV = V\Lambda \Rightarrow \boxed{V^{-1}AV = \Lambda} \quad \boxed{A = V\Lambda V^{-1}}$$

eigenvectors are orthonormal...

Finding a set of basis vectors V such that the original matrix A is diagonal in that basis space, assuming that the columns in V are linearly independent.

Spectral Theorem

Every symmetric matrix has the factorization $S = Q\Lambda Q^{-1}$ with real eigenvalues in Λ and orthonormal eigenvectors in the columns of Q : $S = Q\Lambda Q^{-1} = Q\Lambda Q^T$ with $Q^{-1} = Q^T$.

I.e. Eigenvectors of a real symmetric matrix are always perpendicular.

Proof: Let's assume that x and y are two eigenvectors of S with eigenvalues λ_1 and λ_2 where $\lambda_1 \neq \lambda_2$.

$$Sx = \lambda_1 x,\ Sy = \lambda_2 y$$

$$(Sx)^T y = (\lambda_1 x)^T y,\ x^T S^T y = x^T \lambda_1 y,\ x^T S y = x^T \lambda_1 y$$

$$\therefore \quad x^T \lambda_2 y = x^T \lambda_1 y$$

Since $\lambda_1 \neq \lambda_2$ then $x^T y = 0 \rightarrow x \perp y$.

In class example.

1. $A = \begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix}$

① Is A pos. def?

② Are eigenvectors orthogonal? or orthonormal?

① Find eigenvalues

$$\det\left(\begin{bmatrix} 3-\lambda & 1 \\ 1 & 3-\lambda \end{bmatrix}\right) = 0$$

$$(3-\lambda)(3-\lambda) - 1 = 0$$

$$\lambda^2 - 6\lambda + 8 = 0$$

$$(\lambda - 2)(\lambda - 4) = 0$$

① Yes, because both the eigenvalues are positive

② $\lambda = 4$

$v = N(A - 4I)$

$$\begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix}\begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \xrightarrow{R_2 = R_2 + R_1} \begin{bmatrix} -1 & 1 \\ 0 & 0 \end{bmatrix}\begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$v = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$, $\lambda = 4$

$v_2 = 1 \quad -1v + 1 = 0 \rightarrow v_1 = \underline{1}$

② $\lambda = 2$, $v \in N(A - 2I)$

$$\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}\begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}\begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$v = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$, $\lambda = 2$

$v_2 = 1 \quad v_1 = -1$

Are they orthogonal?

Are they orthonormal?

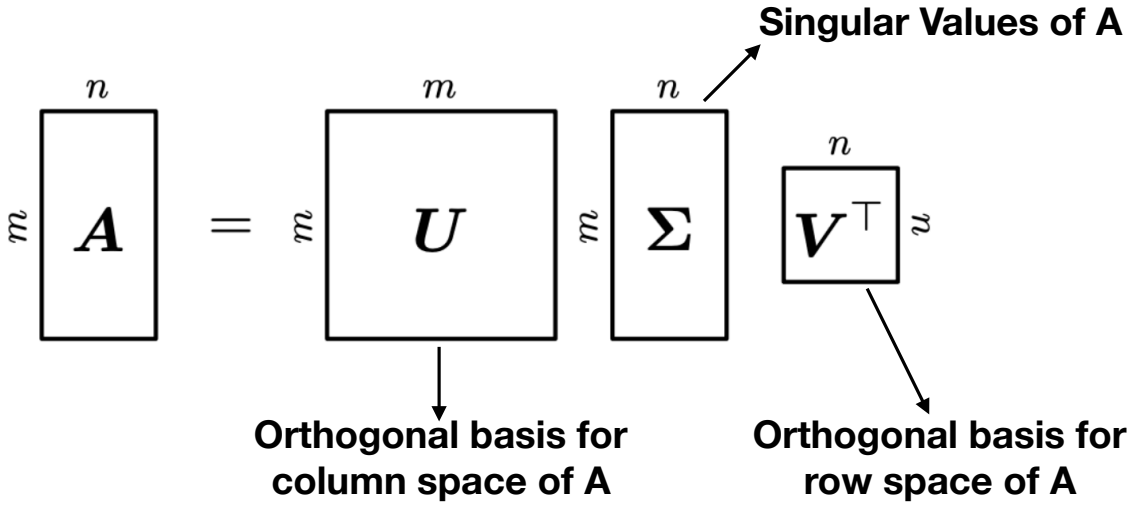
② ↳ No, not size 1

③ $\sqrt{1^2 + 1^2} = \sqrt{2}$

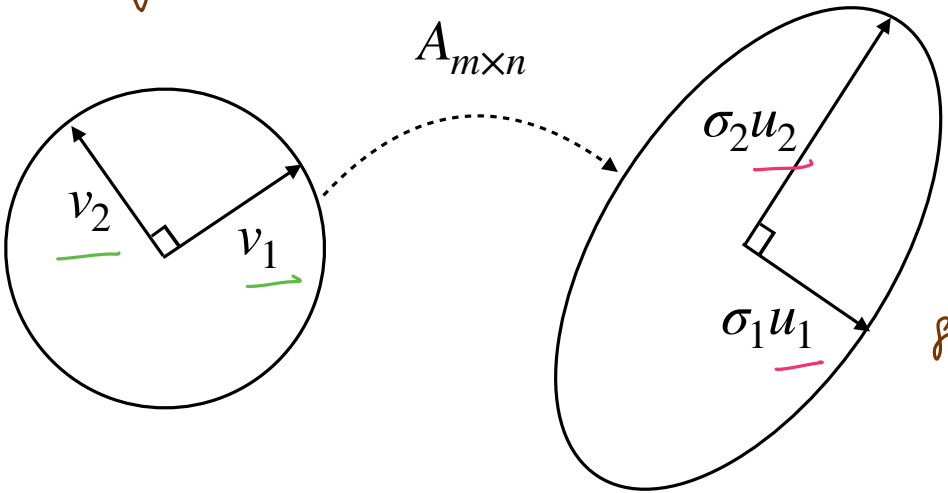
Singular Value Decomposition

SVD

The goal of SVD is to decompose a matrix A as the product of 3 other matrices $A = U\Sigma V^T$, where matrix V and U are orthogonal matrices and Σ is a diagonal matrix.



Formulation



$$\{\vec{v_1}, \vec{v_2}, \ldots, \vec{v_r}\} \rightarrow \{\vec{u_1}, \vec{u_2}, \ldots, \vec{u_r}\}$$

$\vec{u_1}, \ldots, \vec{u_r}$ *orthonormal*

$\sigma_1, \ldots, \sigma_r$ *singular values* (stretch factor)

$$\sigma_1 \geq \sigma_2 \geq \sigma_3 \ldots \geq \sigma_r \geq 0$$

$$r = rank(A)$$

$$A\vec{v_1} = \sigma_1\vec{u_1},\ A\vec{v_2} = \sigma_2\vec{u_2} \rightarrow A\vec{v_j} = \sigma_j\vec{u_j},\ \forall j = 1,2,\ldots,r$$

$$\begin{bmatrix} A \end{bmatrix}\begin{bmatrix} v_1 & v_2 & \ldots & v_r \end{bmatrix} = \begin{bmatrix} u_1 & u_2 & \ldots & u_r \end{bmatrix}\begin{bmatrix} \sigma_1 & & & \\ & \sigma_2 & & \\ & & \ddots & \\ & & & \sigma_r \end{bmatrix}$$

$$AV = U\Sigma$$

$$A = U\Sigma V^{-1} = U\Sigma V^T$$

How to compute SVD?

$$A = U\Sigma V^T$$

square matrix & symmetric matrix. → spectral thm.

$$A^TA = (U\Sigma V^T)^T U\Sigma V^T$$

$$A^TA = V\Sigma^T U^T U\Sigma V^T$$

U is orthogonal

$$A^TA = V\Sigma^T I\Sigma V^T$$

$$A^TA = V\Sigma^T\Sigma V^T$$

- columns of V are eigenvectors A^TA.
- singular values σ are the $\sqrt{\text{eigenvectors.}}$

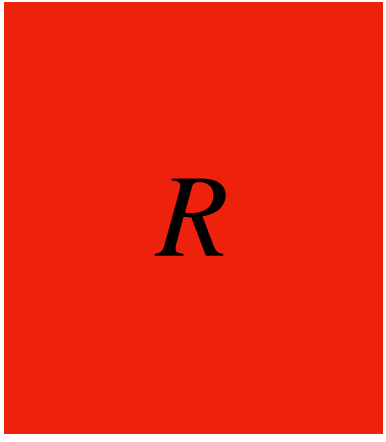
$$A^TA = V\Sigma^2 V^T$$

To find V compute eigendecomposition of A^TA where V will be the eigenvectors of A^TA and Σ^2 are the eigenvalues of A^TA.

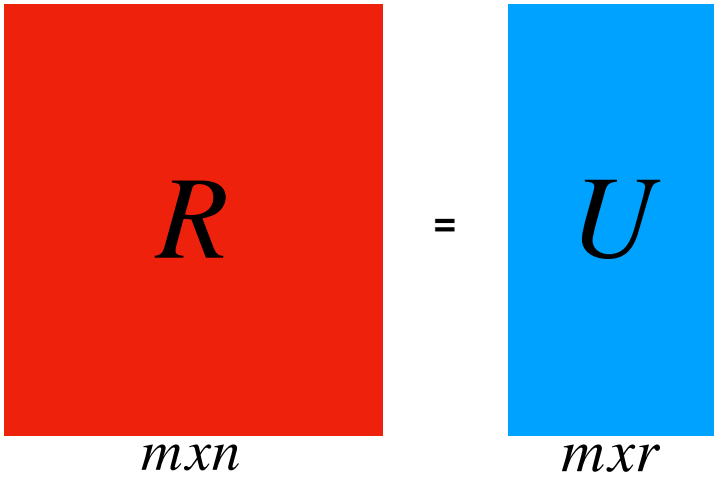
$$AA^T = U\Sigma^2 U^T$$

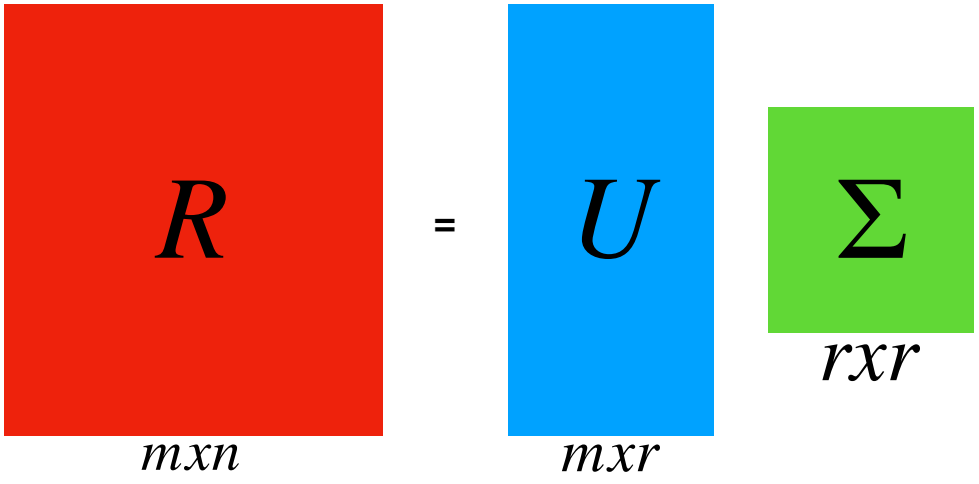
To find U compute eigendecomposition of AA^T where U will be the eigenvectors of AA^T and Σ^2 are the eigenvalues of AA^T.

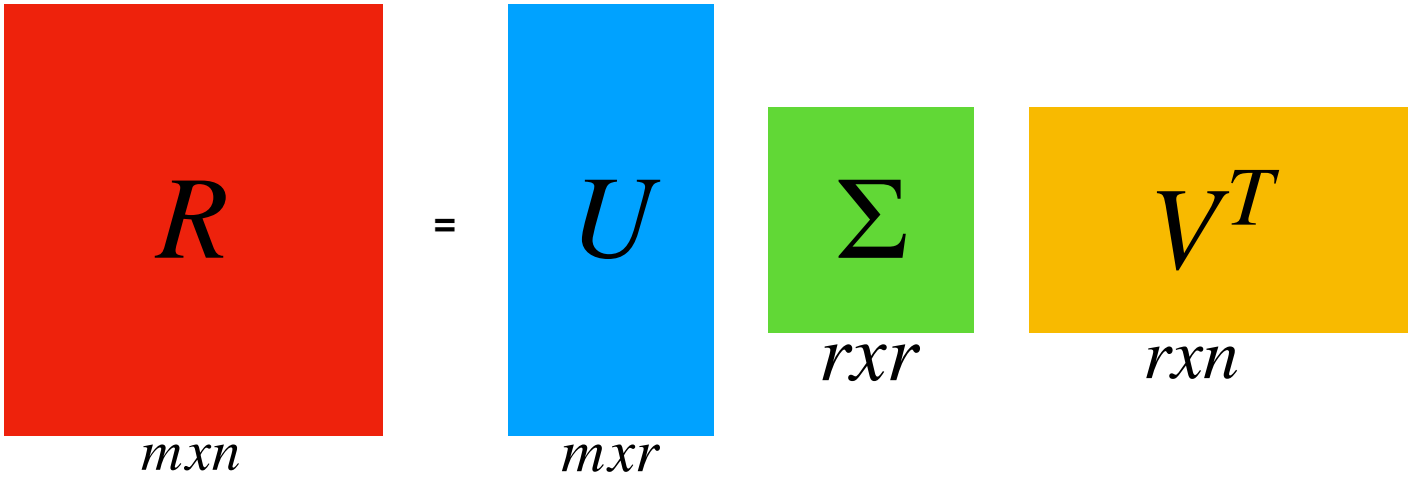
Dimensionality Reduction with SVD

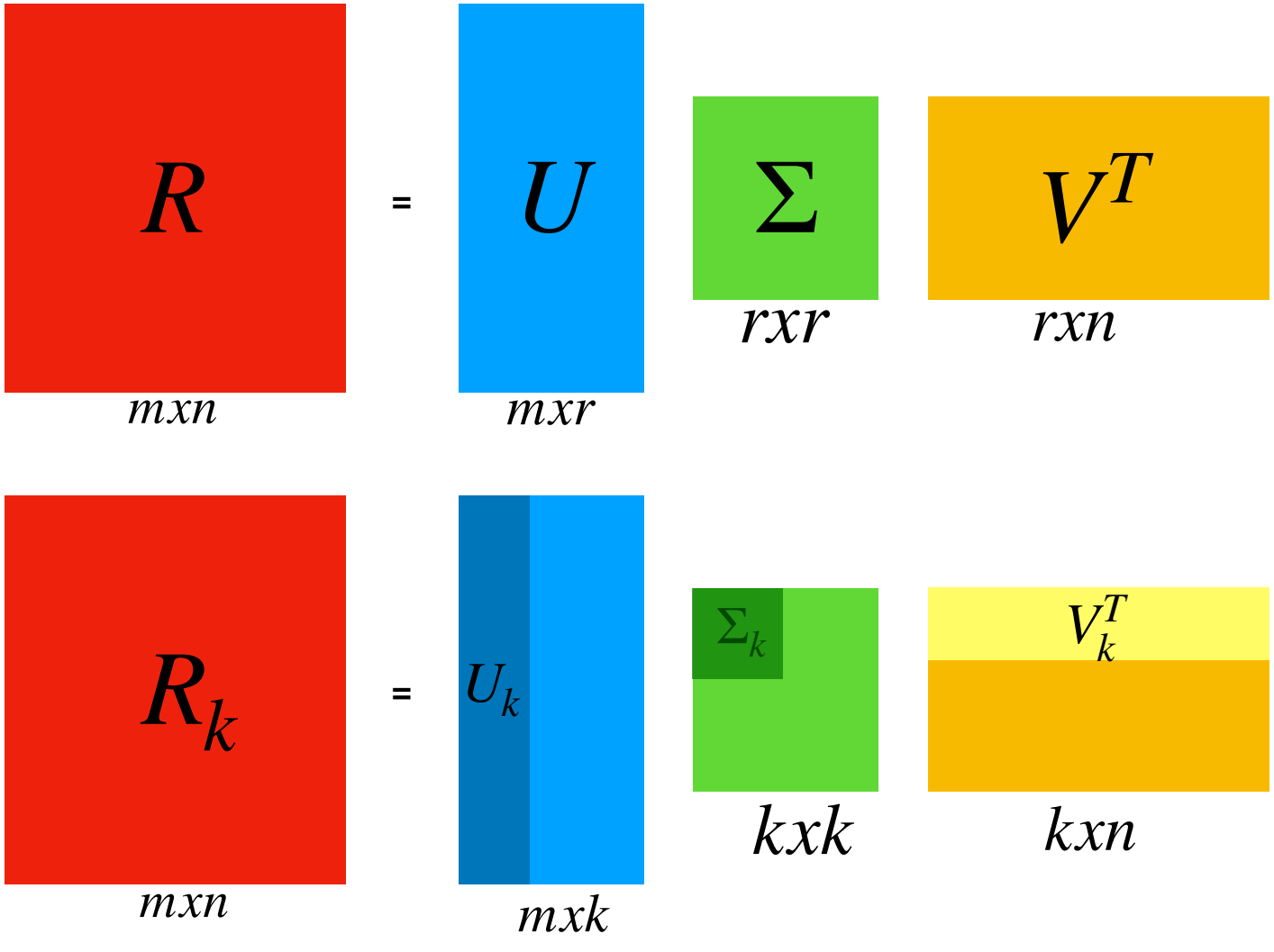


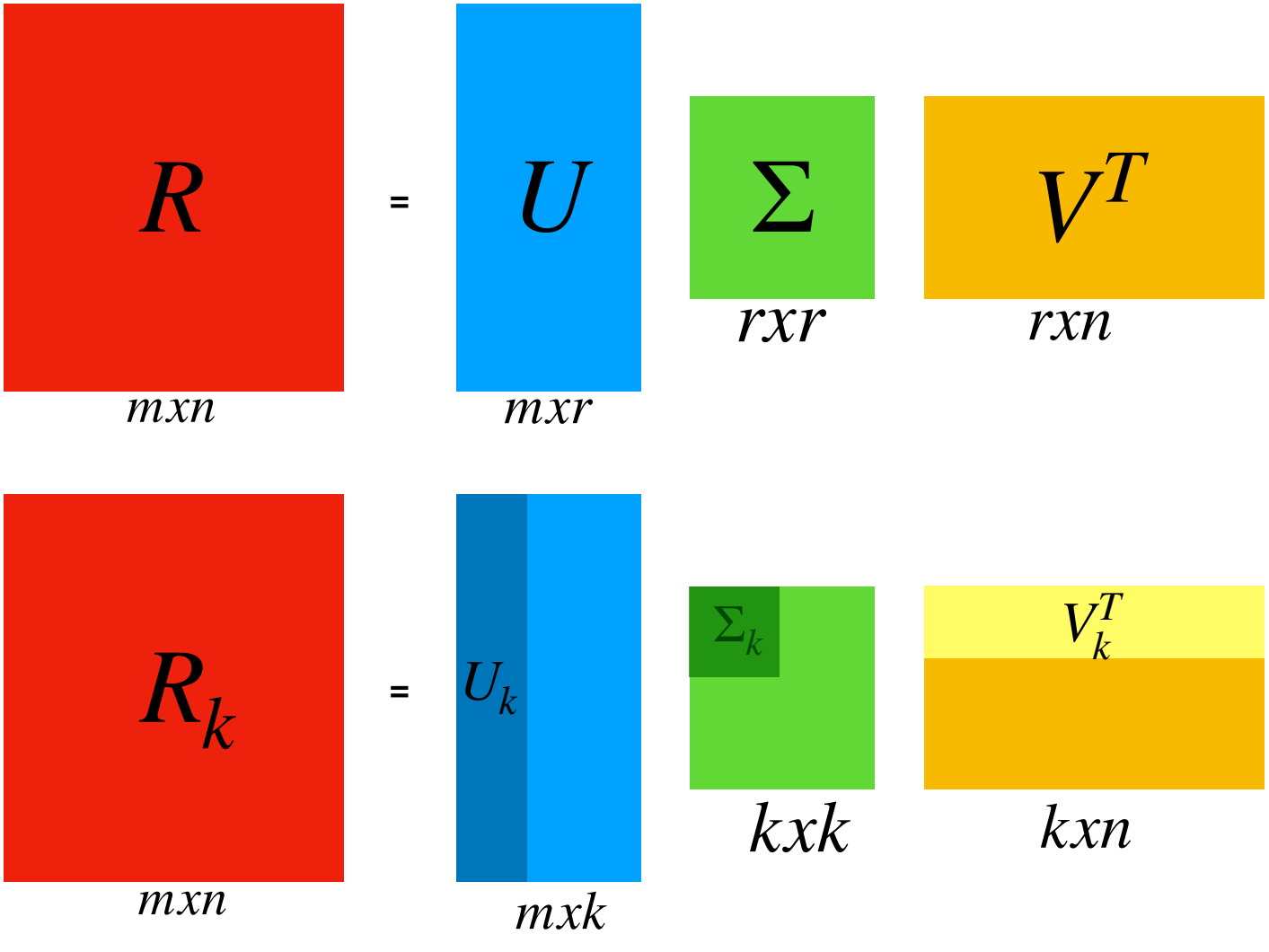
mxn


R
=
U
mxn
mxr


R
=
U
Σ
mxn
mxr
rxr


R
=
U
Σ
V^T
mxn
mxr
rxr
rxn


R
=
U
Σ
V^T
rxr
rxn
mxn
mxr
R_k
=
U_k
Σ_k
V_k^T
kxk
kxn
mxn
mxk



Eckart - Young Theorem 1936 [1]

The reconstruction matrix $R_k = U_k \Sigma_k V_k^T$ is the closest rank-k matrix to R.

Movie Recommendation with SVD

User-Movie Rating









John	5	1	3	5
Tom	?	?	?	2
Alice	4	?	3	?

Image by: Di Noia, Tommaso & Ostuni, Vito. (2015). Recommender Systems and Linked Open Data. 10.1007/978-3-319-21768-0_4.

User Movie Rating

- Overfit representation of user tases

Alice





John



- Computational Complexity, potentially poor results

- Goal: A more compact representation of users tastes and items descriptions

  - Represent our tastes not in terms of the products we like and dislike, but in terms of higher level attributes

How to create these representations?

- Singular Value Decomposition (SVD)
- Intuitive Description: Reduce space to a smaller taste space that is compact and robust

Singular Value Decomposition

$$R = U\Sigma V^T$$

- *R is mxn rating matrix*
- *U is mxk user feature affinity matrix*
- *V is nxk item feature relevance matrix*
- *Σ is kxk diagonal feature weight matrix*
- *Exists for any real R*

Rating Matrix

	The Notebook	Captain America	Seven	Forrest Gump
User 1	5	-	-	5
User 2	-	5	-	3
...				
User m	1	5	5	-

User Feature Matrix

	Drama	Thriller	Action	Fantasy
User 1	0.7	0.2	0.1	0
User 2	0	0.05	0.75	0.2
User m	0.4	0.3	0.2	0.1

Feature Diagonal

	Drama	Thriller	Action	Fantasy
Drama	0.4	0	0	0
Thriller	0	0.3	0	0
Action	0	0	0.27	0
Fantasy	0	0	0	0.05

Movie Feature Matrix

	Drama	Thriller	Action	Fantasy
The notebook	1.0	0	0	0
Captain America	0	0.1	0.7	0.2
Seven	0.5	0.5	0.1	0
Forrest Gump	0.98	0	0	0.03

Rating Matrix

	The Notebook	Captain America	Seven	Forrest Gump
User 1	5	???	-	5
User 2	-	5	-	3
...				
User m	1	5	5	-

User Feature Matrix

	Drama	Thriller	Action	Fantasy
User 1	0.7	0.2	0.1	0
User 2	0	0.05	0.75	0.2
User m	0.4	0.3	0.2	0.1

Feature Diagonal

	Drama	Thriller	Action	Fantasy
Drama	0.4	0	0	0
Thriller	0	0.3	0	0
Action	0	0	0.27	0
Fantasy	0	0	0	0.05

Movie Feature Matrix

	Drama	Thriller	Action	Fantasy
The notebook	1.0	0	0	0
Captain America	0	0.1	0.7	0.9
Seven	0.5	0.5	0.1	0
Forrest Gump	0.98	0	0	0.03

$$R(User_1,\ Captain\ America) = U_{User_1} S V^T_{Captain\ America}$$

References

[1] https://citeseer.ist.psu.edu/viewdoc/download;jsessionid=6B65A9A35513A03050594C88798801DB?doi=10.1.1.23.1831&rep=rep1&type=pdf

Thank you!

In-Class Exercise

Find all eigenvalues and eigenvectors of matrix A.

$$A = \begin{bmatrix} 3 & 1 \\ 4 & 6 \end{bmatrix}$$

In Class Exercise

Compute SVD for matrix A:

$$A = \begin{bmatrix} 4 & 0 \\ 3 & -5 \end{bmatrix}$$